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Patch-Based Breast Cancer Histopathological Image Classification using Deep Learning

Aashita Rajput, Muskan Yadav, Sachin Yadav, Megha Chhabra*, and Arun Prakash Agarwal

Department of Computer Science and Application, School of Engineering and Technology, Sharda University, Greater Noida, India

Abstract

One of the significant contributors to the present death rate for women is breast cancer. An earlier diagnosis of this illness can save costs and increase survival rates. In the past, machine learning algorithms that were used to categorize data were only dependent on manually created features, which could not adequately capture variations and needed a higher degree of classification accuracy. The classification problem was also solved using deep learning methods. The experiment seeks to categorize the cancerous cell types — benign and malignant - using a convolutional neural network (CNN) based model. It is the most incredible technique for classifying images since it is one of the best at identifying the items and patterns in an image. It is a that have a kind of deep learning approach that is best known for spotting patterns in images; it aims to present images in an abstract form that contains the most soundless information required for differentiating them from other images with a similar appearance. BreakHis, a dataset with microscopic biopsy images, and BUSI, a dataset with histopathology images, were employed for this investigation. The primary goal of this experiment is to experiment with the model used to diagnose breast cancer by categorizing various breast photos as either cancerous (Malignant) or non-cancerous (Benign).

Keywords: neural networks; patch-based classification; deep learning; transfer learning; breast cancer

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1. Introduction

Years according to figures compiled over several years, breast cancer affects more than 3 million women in the United States. It starts spreading from the milk Duct and eventually can spread up to fatty tissue outside the milk Duct or even in lymph nodes, reducing the survival rate to 89%. Detection of IDC is time-consuming as this pathologist visually inspects the sample using a microscope [1]. It would be very beneficial for pathologists if this process is automated, so this has been a matter of great interest for of researchers.

Several factors are multi-functional and responsible for breast cancer [2]. The following factors are family, diet, hormones, obesity, history, reproduction of abnormal cells, and radiation therapies. Stage grouping is a technique used to combine data from a patient's T, N, and M categories to determine the stage of a woman's disease [3]. This progression is shown as a transition from the initial stage to the most advanced stage, which is from Stages 0 to 4. It is a malignant tumor that that forms from breast cells. The term "breast cancer" refers to the spite of breast tissue. It is the most common type of cancer in women worldwide, affecting 10% of all women in the Western world at some point in their lives. [4]. Despite the importance of breast cancer, researchers still don't fully understand the causes of the majority of cases. Still, they do understand some risk factors (such as aging, genetic risk factors, family medical history, menstrual cycles, and not having children) that increase a woman's chance of developing breast cancer [5].

The computer-aided diagnostic (CAD) system is a second option to cure multi-classification difficulties for cancer. A CAD system provides a low-cost, easily accessible, quick, higher-quality design with documentation and a dependable source of diagnosis. This system has reduced the death rates by 30 to 70%. Such images are used in machine learning categorization as diagnostic images. Artificial intelligence has a component called machine learning [6].

* Corresponding author.
E-mail address: Megha.chhbr@gmail.com

Machine learning to the method of collecting meaningful knowledge from massive volumes of data by employing algorithms to search for patterns and learn in a prescribed manner. ML algorithms, to instead of depending on any pre-existing equation as a model, use computational approaches to learn directly from data [7]. utilized one of three methods using machine learning can be used to train the model. Supervised machine learning uses known data and is controlled by a supervisor. When there is no supervision, machine learning takes place unsupervised. Less often, supervised learning is applied in machine learning [8]. These algorithms use the best data from prior knowledge to make the best possible decisions.

The main contributions of the proposed work are:

- Full breast cancer image is divided into equal-sized patches P x P. The patch sizes experiment with P= {8,16,32,50}.
- Improvements in CNN are proposed to classify breast cancer using image patches Batch normalization and Dropout layer with the amounts to {No Dropout, 0.1 and 0.2}.
- Evaluations of the CNN model against pre-trained models are performed to evaluate the effectiveness of the proposed CNN work.

The following paper consists of a section named Related Work, which includes a detailed review of all the related work done in this domain, followed by the proposed work, which provides details on the experimentation performed. The section named Experimentation Results includes all the related performance evaluation tables and graphs for all the experimented models.

2. Related Work

Deep learning techniques used in machine learning are comparable to how some types of knowledge are learned by humans. Deep learning is a crucial aspect of data science, encompasses statistics and forecasting. Deep learning makes this process faster and simpler for data scientists tasked with gathering, interpreting, and condensing a significant amount of data [9]. The deep learning algorithm consists of four steps: (1) collecting data, (2) selecting the optimal neural network, (3) training the neural network using input data and parameters, and (4) evaluating the neural network using test data.

Work in [10][11] involving unsupervised descriptor extraction and deep learning with convolutional neural networks prepared the path for an autonomous representation technique. H. notwithstanding any potential human interference that might impact these performances. To train effectively, deep CNNs frequently include large number of parameters. The size of medical imaging datasets is insufficient for adequate teaching a deep CNN model. Thus, research in the literature [12] examined the utilization of transfer learning in medical image processing. Transfer learning aims to move knowledge from a broad source domain to a narrow target area. For CNNs, this is often done by pre-training a CNN model using the source dataset and then retraining part of the model using the target dataset [13].

Most techniques (based on textural, form, marginal, and multiresolution research) only extract low- or middle-level characteristics, failing to significantly advance the categorization process. To remove information from the mammograms, Arevalo et al. used CNN [14]. A support vector machine was then used to finish the categorizing process. F. Along with L.S. Oliveira [15] used the Support Vector Machine (SVM) classification approach to divide an image into two categories after extracting the properties of the image using either Gray Level Co-Occurrence Matrices (GLCM) or Local Binary Patterns (LBP).

Yap et al. [16] created and used two datasets based on ultrasound images and exhibited the most outstanding performance of a transfer-learning FCN-AlexNet approach. Many architectures generated comparable results, including patch-based and the U-Net approach. Their findings were compared to those of various CNN modelling techniques (DPM) and many other deep learning techniques. On the contrary, he expressed the need for such tactics for a long training duration as well as negative impressions.

Byra et al. [17] used the technique of Nakagami parameter maps to build a model based on CNN, including an area underneath the ROC curve (AUC) of approximately 0.9. In the experimentation conducted, they used and experimented with 458 RF ultrasound images. They observed that quantitative ultrasound factors (QUS) can influence the Nakagami technique and that CNN classification is directly related to image quality.

Even though CNNs were used by Hepsag et al. [18] on different datasets. Performance ratings for basic studies ranged from 60 to 72%. The accuracy of the BCDR data sets improved from 65 to 85% when pre-processing techniques were used. Similar gains were observed in the mini-MIAS data set.

Abdel-Zaher and Eldeib [19] used a deep CNN to classify breast cytology images (DBN). They were 99.68% accurate, 100% sensitive, and 99% specific. They employed DBN unsupervised to recognize characteristics within the Wisconsin Breast Cancer Dataset (WBCD). A supervised phase was also constructed using deep backpropagation, conjugate gradient architecture, and Levenberg-Marquardt optimization methods. Using the conjugate gradient and Levenberg-Marquardt algorithms, the authors claim that their proposed network model (DBN-NN) accuracy is better than that of an arbitrarily set base weight backpropagation NN (RIW-BPNN).

Hadad et al. investigated three CNN models. [20] for MRI breast lesion classification, the authors' accuracy rates were 0.94, 0.90, and 0.93. Mammography images (282 increased to 32,064) and MRI images were added by the authors (123 increased to 19,316). They started by employing a CNN trained from scratch using MR images. Second, they used VGG-Net to put a fine-tuned CNN to the test. They eventually used MG-Net to create a fine-tuned CNN.

3. Methodology

For the proposed work, the following steps were undertaken.

- First, each image is gathered from the BreakHis and BUSI datasets, which contain histopathology images of benign and malignant breast cancer.
- These images are then resized into a consistent size of 400x400 and further divided into patches, each of size PxP, where P varies as 8, 16, 32, and 50.
- These patches are later classified into benign and malignant using CNN-based architectures, which includes the experimentation with the dropout layer with varies from no dropout used, with values as 0.1 and 0.2, and the use of pre-trained models like VGG, DenseNet, and ResNet.
- The architecture's performance over these patches is evaluated using various performance metrics parameters like accuracy, loss, F1- score, and ROC, later, in terms of summarization, the cumulative ROC for each architecture is also drawn.

The complete architecture adopted for this work is diagrammatically depicted in Figure 2. Along with this, the various outcomes achieved at each stage of the proposed work are as follows:

1) Data Collection

The first step in building the system will be to gather a large dataset of breast cancer images and corresponding labels (i.e., benign or malignant). This dataset will be used to train and test the neural network [21]. The distribution of the training and testing dataset is provided in Table 1.

2) Image Pre-Processing

Before the images can be fed into the neural network, they will need to be pre-processed to ensure that they are in a format that the network can understand [22]. The proposed system along with resizing the image includes patch conversion. The outcome of image pre-processing is shown in Figure 1.

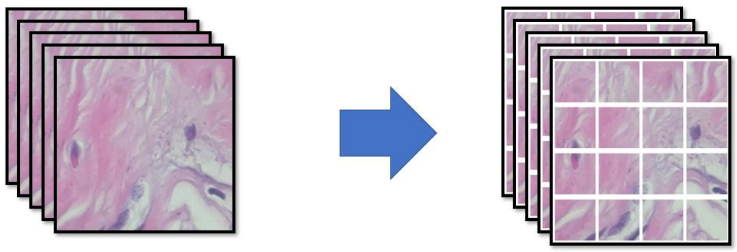


Figure 1. Image Pre-Processing

4. Deep Learning Architectures

1) Convolutional Neural Network

These neural network designs make use of the following three ideas: pooling, extensive weight sharing, and local receptive fields [23]. The region (local area) in which the filter is applied is known as the local receptive field. A CNN consists of many different types of layers, including a convolutional layer, a pooling layer, and a fully connected layer. These levels are used to build the entire CNN architecture [24].

- *Convolutional Layer*

Shift, multiply, and sum operations are used in CNN's convolutional layer, which is its core algorithm. Its parameters are made up of kernels, which are customizable filters or weight matrices [25]. The convolutional layer's purpose is to find patterns in the small portions of the input images that are shared by the entire dataset. Convolution is accomplished by multiplying the receptive parameter value with the filter one at a time [26]. To introduce non-linearity, the weighted total is added with a bias and then passed via an activation function. The filter then advances through the number of iterations indicated by the stride variable. The same technique is repeated using the identical filter to achieve the value indicated below. The same technique is repeated using the identical filter to achieve the value indicated below [27]. This results in the creation of a feature map for each filter in each convolutional layer [28]. The convolution process is described as follows for one-dimensional data and five local connection networks:

$$V_j = \sum_{i=1}^{l} w_i x_{i+j-1} \; j = 1,2,3,4,5 \tag{1}$$

where w denotes the shared weight matrix containing all hidden neurons and l denotes the kernel size. Because Equation 1 above is written as a convolution sum, a network built in this manner is known as a convolutional neural network [29]. The feature map presented below was created using the previously mentioned Equation 1. The inclusion of a bias term follows the usage of a non-linear function is shown in Equation 2:

$$h_k = \sigma(V_j^k + b_k) \tag{2}$$

Where b_k is the bias term, h_k is the $k - th$ feature map and σ is the non-linearity function.

- *Pooling Layer*

Each convolutional layer is followed by another pooling layer that facilitates the retention of information in the output of the convolutional layer [30]. A pooling layer summarises the arrangement of neurons in the convolutional layer by sampling each feature map produced by the convolutional layer. Lowering the geographic scale of the input lowers the computing needs and network parameters, which is the fundamental purpose of the pooling layer [31]. Max Pooling and Mean Pooling are two of the most popular pooling techniques among the numerous available. Simply put, max pooling returns the maximum value of the input region. The input area is a variable-sized chunk of incoming data. The mean value of all the numbers in the input region is what results from pooling the mean value [32].

- *Fully Connected Layer*

Convolutional layers are followed by fully connected layers in a typical CNN architecture. Each neuron in a fully connected layer is connected to each neuron in the next layer, and each number receives a vote indicating how much it resembles that class [33].

2) VGG

Visual Geometry Group, or VGG, is a common multi-level deep convolutional neural network (CNN) architecture. "Deep" refers to the total number of layers in VGG-16 and VGG-19, which respectively contain 16 and 19 convolutional layers [34]. The VGG architecture is the basis of ground-breaking object recognition models. Built as a deep neural network, VGGNet outperforms baselines for various tasks and datasets in addition to ImageNet. Moreover, it is still one of the most widely used image recognition architectures [35].

3) ResNet

The ResNet model is one of the most popular and successful deep learning models. These Residual blocks have made it possible to train incredibly deep networks, and the ResNet model comprises these blocks [36].

4) DenseNet

A convolutional neural network known as DenseNet uses dense blocks to directly connect all layers (each with corresponding feature map sizes) to take advantage of dense connections between layers. To maintain the architecture moving forward, each layer communicates its own feature maps to every subsequent layer and gets new information from all prior layers [37].

5. Performance Analysis

The various performance analysis measures used in this research are accuracy, ROC, and F1-score.

- *Accuracy*

One crucial measure to consider while assessing a classifier's performance is classification accuracy [36]. This measure demonstrates the degree of data categorization accuracy for a given classification model, which is represented as Equation 3.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{3}$$

- *ROC*

A graph that shows how well a classification system performs at all classification levels is called a receiver operating characteristic curve (ROC curve) [37]. On this graph, two metrics—True Positive Rate (TPR) and False Positive Rate—are displayed (FPR).

- *F1–Score*

A statistic known as the F1-score can be used to determine the equilibrium between recall and precision. By taking into consideration both recall and precision rates, the F1 score provides a practical way to evaluate how accurate the model is [38]. The Equation 4 yields the F1 score:

$$F1 - score = 2 \times \frac{Recall \times precision}{Recall + precision} \tag{4}$$

6. Proposed Architecture

The suggested classification method separates the input image into patches of different sizes like 8x8, 16x16, 32x32, and 50x50 patches, which are then categorized into cancerous and benign cells using a CNN model called Patch-wise Classifier. Figure 2 displays the suggested model's block diagram.

The CNN has five convolutional layers with weights, two layers of maximum pooling, three layers of density, one layer of flattening, and one layer of dropout. The first convolutional layer applies 64x3x3 filters with padding to the 50x50 input picture block. Rectified Linear Units (ReLu) layers are placed after each convolution layer to add nonlinearity. The maximum pooling layer follows the second convolutional layer. The combined output from the second convolutional layer is reused as the source for the third convolutional layer, which filters it using 128 filter sizes that are 3x3x64.

Using 128 kernels with the size of 3x3x64, the fourth convolutional layer filters the input from the previous convolutional layer. The fifth layer of the architecture filters the output with 256 filters of size 3x3x64. The output is then transmitted to another pooling layer as the max pooling layer using a pool size of 2x2, followed by the fifth convolutional layer. The fifth convolutional layer's output is then flattened, a dense of unit 128 with ReLu activation is applied, and a dropout of 0.15 is then applied [38]. Ultimately, the network has a completely integrated layer that produces binary classification using a SoftMax classifier.

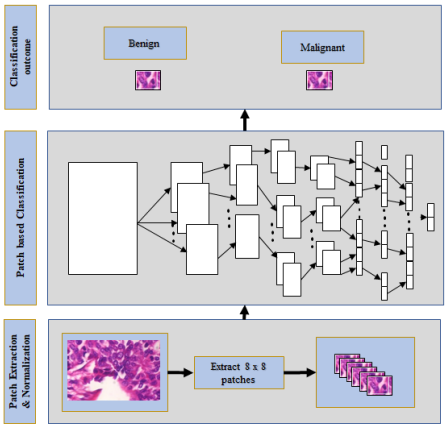


Figure 2. Systematic Architecture

7. Experimental Results

The study used two approaches to image pre-processing and trained the network using the Adam optimization technique. The model was trained in 100 epochs and the performance was analysed in terms of accuracy, loss, f1-score, and ROC using the test and train datasets as shown in Table 1. These datasets were resized into the constant size of 400 x 400 and subdivided into patches of size 8x8, 16x16, 32x32, and 50x50. Table 1 shows the distribution of the training and testing dataset where each image patch is of size 50 x 50.

Table 1. Distribution of Train and Test Datasets

Dataset	Train and Test	Label	
		Negative	Positive
BreakHis400X	Train	5936	12,432
	Test	2816	5904
BUSI	Train	3488	1680
	Test	3488	1680

The evaluated f1-score and ROC values presented in the study provide a quantitative measure of the model's performance, which can be useful for assessing its potential clinical utility.

7.1 BreakHis400X

The following are the results achieved from the dataset named “BreakHis400X” with a significant number of positive and negative test and train data as described in Table 1. It contains the microscopic biopsy images of benign and malignant breast tumors, it includes only the 400x optically zoomed images, specifically provided for deep learning models, and the images present are in PNG format. Figure 3 depicts both the samples for the benign as well as the malignant tissues.

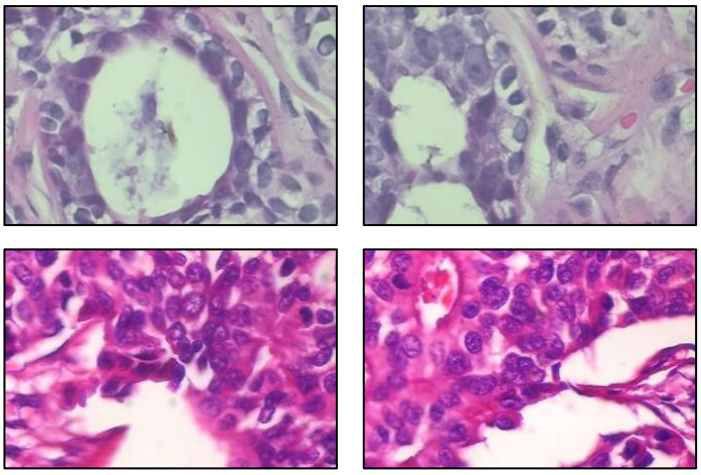


Figure 3. BreakHis400X dataset image samples

Each image from the dataset is then pre–processed to generate the patches each of size 8x8, 16x16, 32x32, and 50x50 as the sample shown in Figure 4. These patches are locally stored in the system and passed into the algorithm as training and testing samples. The CNN-based outcome on this dataset is shown in Figure 5. The figure displays the Training and Validation Accuracy and Training and Validation Loss along with the ROC curve. Figure 6 shows the cumulative ROC curve for CNN, VGG16, ResNet, and DenseNet121.

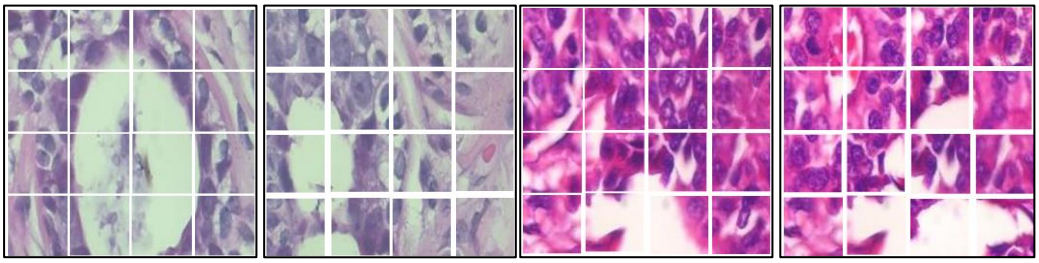


Figure 4. BreakHis400X dataset images patches

7.2 BUSI

The following are the results achieved from the dataset named "BUSI" with a significant number of positive as well as negative test and train data as described in Table 1. The dataset images are the breast ultrasound images with sample images shown in Figure 5, these images were collected among women ages 25 to 75 in the year 2018. It includes images from 750 women, each image's average size is of 400x400 in PNG format.

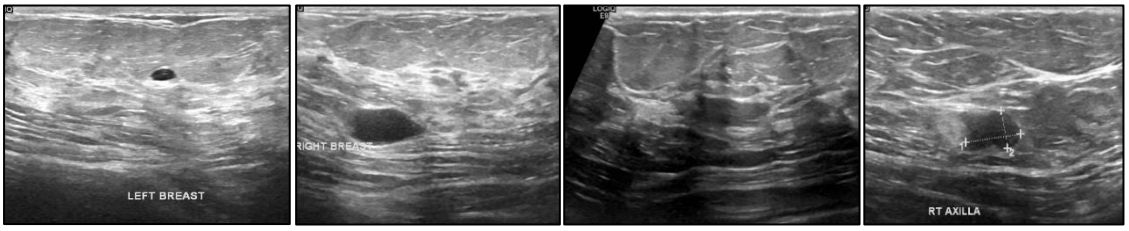


Figure 5. BUSI dataset image samples

Each image from the dataset is then pre–processed to generate the patches of size 8x8, 16x16, 32x32 and 50x50 as the sample shown in Figure 8. These patches are locally stored in the system and passed into the algorithm as training and testing samples. The CNN-based outcome on this dataset is shown in Figure 6. The figure displays the Training and Validation Accuracy and Training and Validation Loss along with the ROC curve. Figure 10 shows the cumulative ROC curve for CNN, VGG16, ResNet, and DenseNet121.

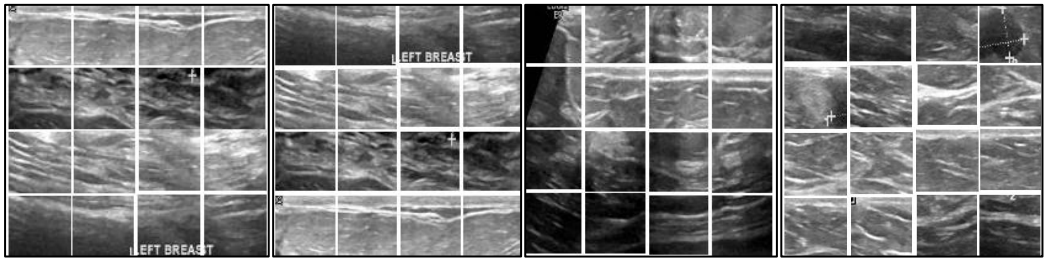


Figure 6. BUSI dataset images patches

7.3 VGG16

The pre-processed image patches are passed through various architectures out of them one is VGG16, the VGG16 architecture used consists of 16 Convolution Neural Network based layers. Along with the basic VGG16 architecture, the experimented model consisted of a global max pooling layer and a dropout of 0.5. The architecture was used to classify breast cancer images gathered from two datasets as depicted in Figures 7 and 8. The accuracies achieved with the help of this architecture are represented with the help of Tables 2 and 3. Table 2 represents accuracies from the BreakHis dataset and Table 3 from the BUSI dataset.

Table 2. VGG16 Architecture Performance Evaluation on BreakHis Dataset

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	64	56	62	63
16x16	65	53	67	67
32x32	71	50	70	70
50x50	84	46	76	74

Table 3. VGG16 Architecture Performance Evaluation on BUSI Dataset

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	65	55	67	68
16x16	68	52	66	66
32x32	69	53	68	70
50x50	74	50	74	72

The above-mentioned tables can be summarised to attain the below-drawn graphs which depict the ROC curves drawn from the false positive rate and true positive rate from the algorithm's performance. Table 2 can be summarized as shown in Figure 7 and Table 3 as shown in Figure 8.

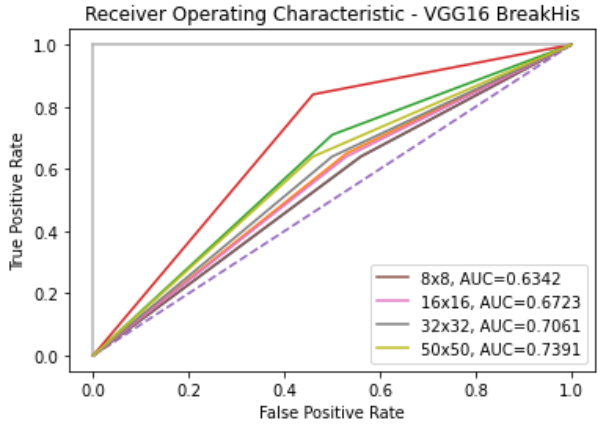


Figure 7. Cumulative ROC for all the patch sizes of BreakHis dataset on VGG16 architecture

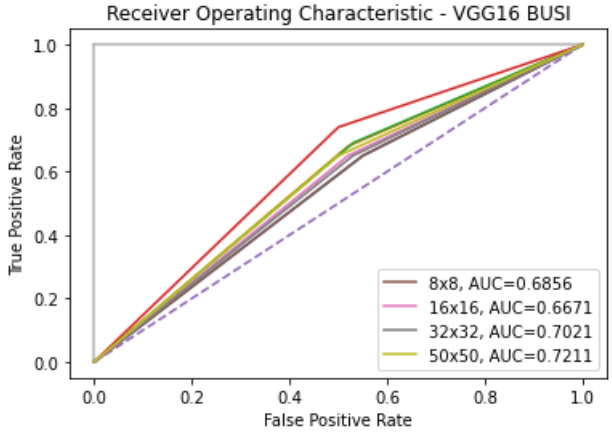


Figure 8. Cumulative ROC for all the patch sizes of BUSI dataset on VGG16 architecture

7.4 ResNet50

Further in the experimentation, the images from both datasets are passed through the ResNet50 model. The model used is ResNet50 which consists of 50 layers of classical CNN architecture. The model was experimented with two datasets – BreakHis and BUSI, these dataset images were converted into patches, each of size 8x8, 16x16, 32x32, and 50x50. The accuracy achieved on the BreakHis dataset is shown in Table 4 whereas the accuracy achieved on the BUSI dataset is shown in Table 5.

Table 4. ResNet50 Architecture Performance Evaluation on BreakHis Dataset

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	68	58	69	67
16x16	70	56	65	69
32x32	73	54	71	71
50x50	80	51	75	77

Table 5. ResNet50 Architecture Performance Evaluation on BUSI Dataset

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	61	51	60	62
16x16	65	54	67	67
32x32	67	58	70	65
50x50	73	60	72	73

Figures 9 and 10 are used to summarize the results of the performance evaluations on different datasets as shown in Tables 4 and 5 respectively. The below-shown figures contain the cumulative ROC curve for the model on a specific dataset. Figure 9 shows the ROC curve of the ResNet50 model on the BreakHis dataset and Figure 10 shows the ROC curve of the ResNet50 model on the BUSI dataset on different patch sizes.

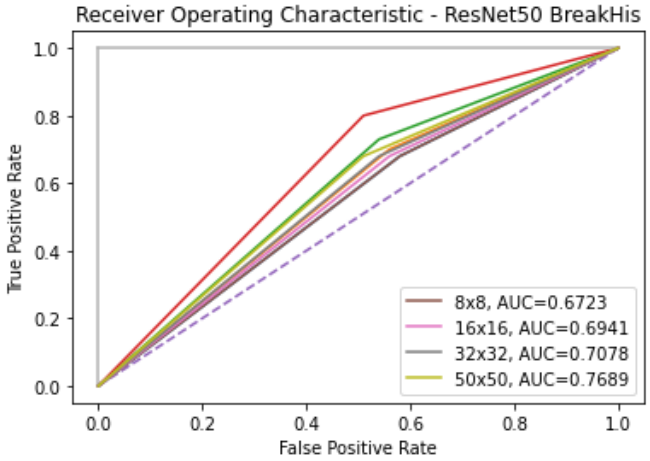


Figure 9. Cumulative ROC for all the patch sizes of BreakHis dataset on ResNet50 architecture

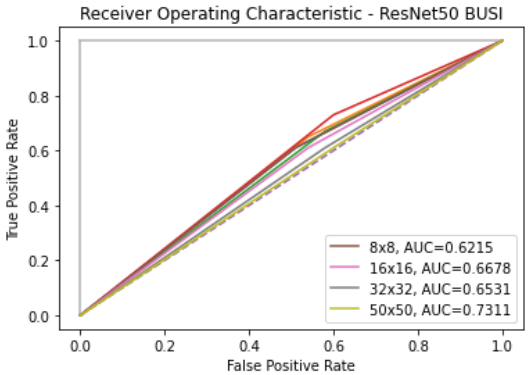


Figure 10. Cumulative ROC for all the patch sizes of BUSI dataset on ResNet50 architecture

7.5 DenseNet121

The experimentation further involved using another CNN-based pre-trained model called DenseNet121. The DenseNet121 architecture used for this experimentation is DenseNet121 which consists of 120 convolutional and 4 average pooling layers. The model further involved using a global max pooling layer and a dropout of 0.5. The accuracy achieved on the BreakHis dataset is shown in Table 6 whereas the accuracy achieved on BUSI dataset is shown in Table 7.

Table 6. DenseNet121 Architecture Performance Evaluation on BreakHis Dataset

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	67	59	65	68
16x16	71	56	73	70
32x32	73	53	76	72
50x50	75	52	84	83

Table 7. DenseNet121 Architecture Performance Evaluation on BUSI Dataset

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	63	58	65	64
16x16	65	56	66	66
32x32	68	53	69	67
50x50	71	51	70	71

The drawn figures, Figures 11 and 12 represent the cumulative summary of DenseNet121 architecture over both datasets. It comprises of the ROC curve drawn from the values given in Tables 6 and 7. Figure 11 represents the cumulative ROC of DenseNet121 architecture over the BreakHis dataset and Figure 12 represents the cumulative ROC of DenseNet121 architecture over the BUSI dataset.

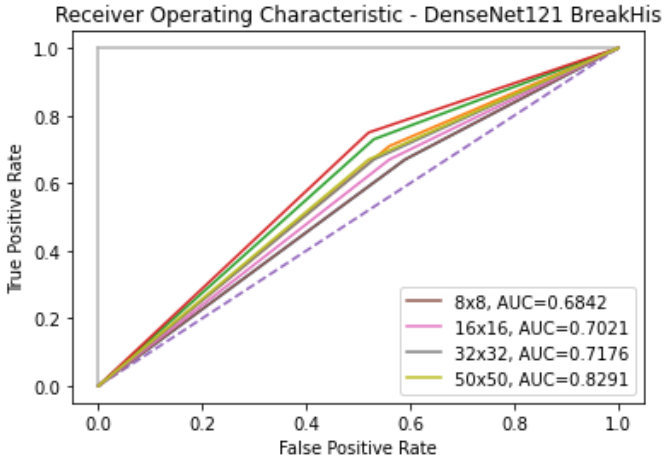


Figure 11. Cumulative ROC for all the patch sizes of BreakHis dataset on DenseNet121 architecture

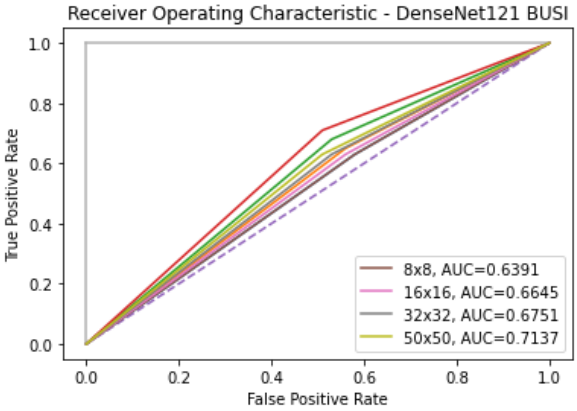


Figure 12. Cumulative ROC for all the patch sizes of BUSI dataset on DenseNet121 architecture

7.6 CNN

The primary experimentation revolves around the CNN model developed using various convolutional neural network-based layers. This experimentation involves experimenting with the dropout value and observing the performance obtained by the model. The dropout values ranged from 0.2, 0.1, and 0 i.e., no dropout was used. This section involves three sub-sections for these dropout values and their performance evaluations.

The CNN architecture design can be classified into three categories. These are CNN with dropout values of 0.1, 0.2, and no dropout. The following Figures 13 to 15 provide a comparative graph of accuracy achieved on these dropout rates over different patch sizes.

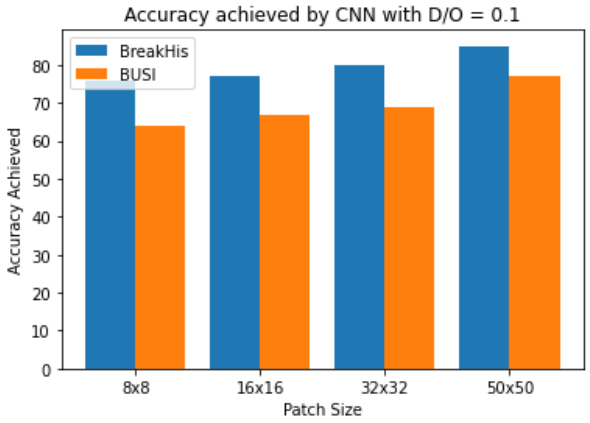


Figure 13. Cumulative Accuracy for all the patch sizes of both datasets on CNN with D/O = 0.1 architecture

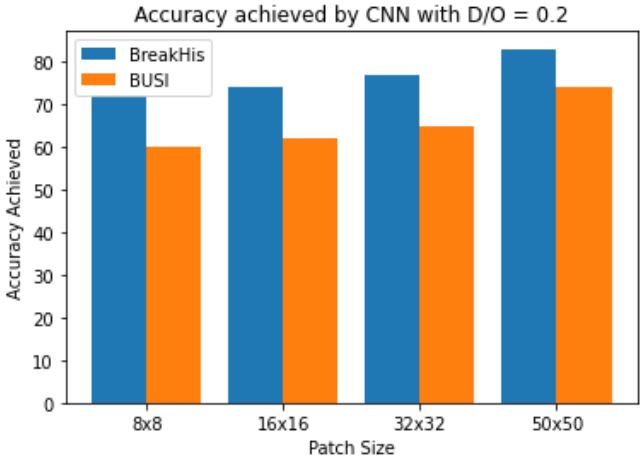


Figure 14. Cumulative Accuracy for all the patch sizes of both datasets on CNN with D/O = 0.2 architecture

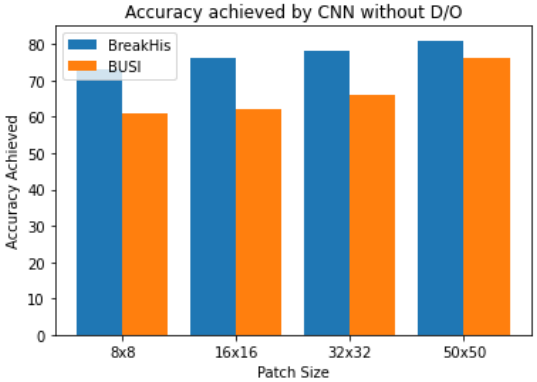


Figure 15. Cumulative Accuracy for all the patch sizes of both datasets on CNN without D/O architecture

7.7 CNN with Dropout value as 0.1

The dropout within a certain range helps generate higher accuracy, helps reduce the complexity of the network, and prevents the overfitting of the output. The following two tables, Tables 8 and 9 depict the performance evaluation of the CNN model with a dropout of 0.1 over two datasets namely BreakHis and BUSI.

Table 8. CNN Architecture Performance Evaluation on BreakHis Dataset with D/O = 0.1

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	76	50	75	71
16x16	77	46	77	76
32x32	80	42	79	75
50x50	85	39	85	86

Table 9. CNN Architecture Performance Evaluation on BUSI Dataset with D/O = 0.1

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	64	60	62	61
16x16	67	59	65	64
32x32	69	56	68	68
50x50	77	52	76	77

Figures 16 and 17 are used to summarize the results of the performance evaluations on different datasets as shown in Tables 8 and 9 respectively. The below-shown figures contain the cumulative ROC curve for the model on a specific dataset. Figure 16 shows the ROC curve of CNN architecture with a dropout of 0.1 model on the BreakHis dataset and Figure 17 shows the ROC curve of CNN architecture with a dropout of 0.1 model on the BUSI dataset on different patch sizes.

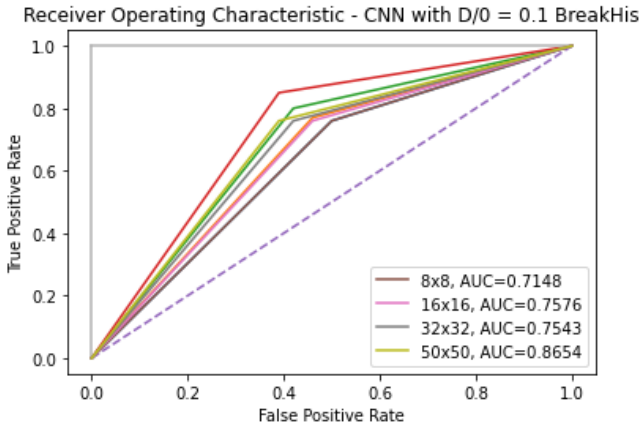


Figure 16. Cumulative ROC for all the patch sizes of BreakHis dataset on CNN with D/O = 0.1 architecture

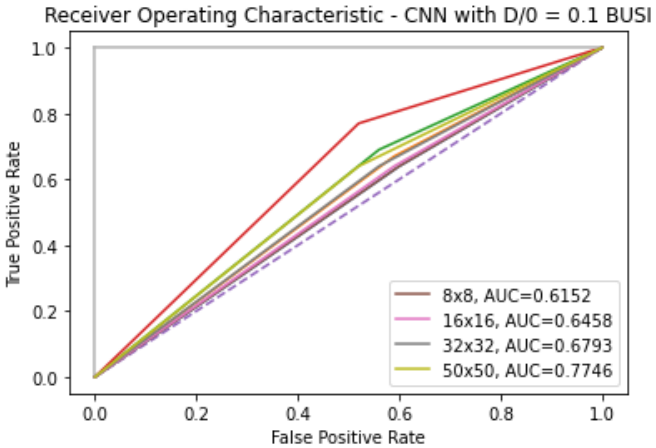


Figure 17. Cumulative ROC for all the patch sizes of BUSI dataset on CNN with D/O = 0.1 architecture

7.8 CNN with Dropout value as 0.2

The following tables, Tables 10 and 11, depict the performance evaluation of the CNN model with a dropout of 0.2 over two datasets namely BreakHis and BUSI.

Table 10. CNN Architecture Performance Evaluation on BreakHis Dataset with D/O = 0.2

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	72	50	77	77
16x16	74	48	79	79
32x32	77	45	81	78
50x50	83	41	82	81

Table 11. CNN Architecture Performance Evaluation on BUSI Dataset with D/O = 0.2

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	60	58	62	62
16x16	62	54	65	60
32x32	65	51	68	65
50x50	74	50	78	75

Figures 18 and 19 summarize the results of the performance evaluations on different datasets as shown in Tables 10 and 11 respectively. The below-shown figures contain the cumulative ROC curve for the model on a specific dataset. Figure 18 shows the ROC curve of CNN architecture with a dropout of 0.2 model on the BreakHis dataset and Figure 19 shows the ROC curve of CNN architecture with a dropout of 0.2 model on the BUSI dataset on different patch sizes.

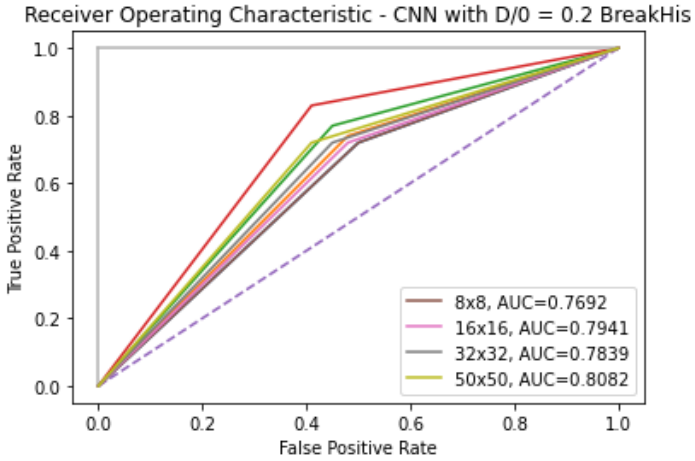


Figure 18. Cumulative ROC for all the patch sizes of BreakHis dataset on CNN with D/O = 0.2 architecture

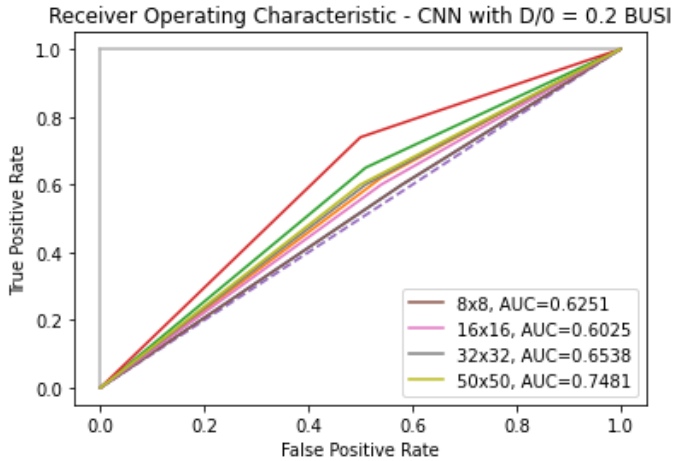


Figure 19. Cumulative ROC for all the patch sizes of BUSI dataset on CNN with D/O = 0.2 architecture

7.9 CNN without Dropout

The following two tables, Tables 12 and 13, depict the performance evaluation of the CNN model without a dropout layer over two datasets namely BreakHis and BUSI.

Table 12. CNN Architecture Performance Evaluation on BreakHis Dataset without D/O

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	73	56	76	72
16x16	76	52	72	78
32x32	78	48	80	77
50x50	81	45	82	78

Table 13. CNN Architecture Performance Evaluation on BUSI Dataset without D/O

Patch Size	Performance Metrics			
	Accuracy (in %)	Loss (in %)	F1-score (in %)	ROC (in %)
8x8	61	59	66	68
16x16	62	58	68	69
32x32	66	55	70	70
50x50	76	53	73	76

Figures 20 and 21 are used to summarize the results of the performance evaluations on different datasets as shown in Tables 12 and 13 respectively. The below-shown figures contain the cumulative ROC curve for the model on a specific dataset. Figure 20 shows the ROC curve of CNN architecture without a dropout model on the BreakHis dataset and Figure 21 shows the ROC curve of CNN architecture without a dropout model on the BUSI dataset on different patch sizes.

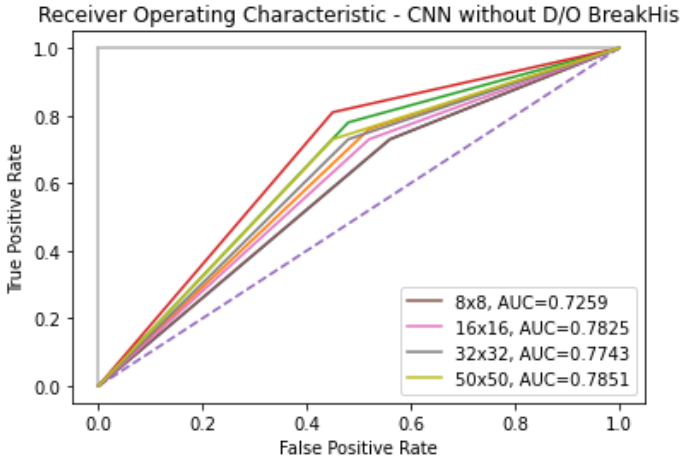


Figure 20. Cumulative ROC for all the patch sizes of BreakHis dataset on CNN without D/O architecture

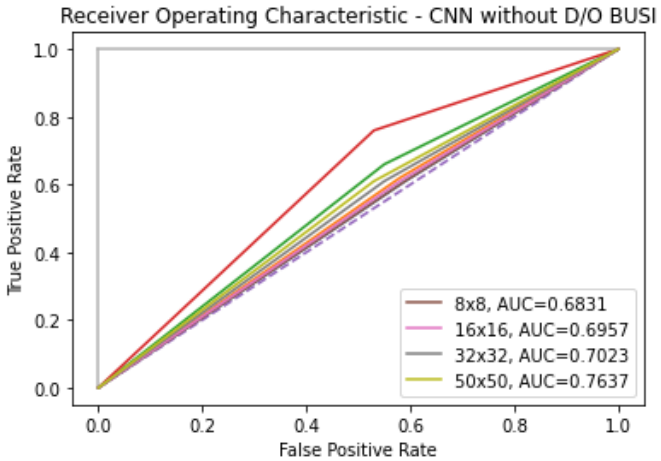


Figure 21. Cumulative ROC for all the patch sizes of BUSI dataset on CNN without D/O architecture

Table 14 contains a cumulative comparison of the accuracies achieved by various techniques over different-sized patches like 8x8, 16x16, 32x32, and 50x50.

Table 14. Accuracy Comparison of Various CNN Techniques

Patch Size	Dataset											
	BreakHis						BUSI					
	VGG (in %)	ResNet (in %)	DenseNet (in %)	CNN (D/O = 0.1) (in %)	CNN (D/O = 0.2) (in %)	CNN (No D/O) (in %)	VGG (in %)	ResNet (in %)	DenseNet (in %)	CNN (D/O = 0.1) (in %)	CNN (D/O = 0.2) (in %)	CNN (No D/O) (in %)
8x8	64	68	67	**76**	72	73	65	61	63	**64**	60	61
16x16	65	70	71	**77**	74	76	68	65	65	**67**	62	63
32x32	71	73	73	**80**	77	78	69	67	68	**69**	65	66
50x50	84	80	75	**85**	83	81	74	73	71	**77**	74	76

Based on the experimentation, the CNN architecture design with a dropout value of 0.1 over a 50x50 patch-sized dataset, provides better accuracy and ROC value as compared to the other techniques like VGG and ResNet whereas DenseNet also provides a better ROC value in comparison to the other two techniques. The developed model worked better with the BreakHis dataset image classification with a dropout value of 0.1 whereas when the dropout value is increased, the model's performance degrades.

8. Discussion

Convolutional Neural Networks can be seen as one of the very crucial deep learning techniques for image classification. Before the experimentation, a comparative study of all the experimentations done by other researchers was taken into consideration and it was concluded that patch–wise image classification can be a significant approach to attain comparatively higher accuracy. The model worked comparatively better on the BreakHis dataset as compared to the BUSI dataset i.e., the histopathological images turned out to be an ideal set of images for breast cancer classification as compared to the ultrasound images.

The proposed model showed comparatively better results over the histopathological images taken from the dataset named BreakHis than the images from the ultrasound-based dataset named BUSI. The model can be further improved through the addition of more layers and the application of feature selection on images. The experimentation can also include using various other pre-trained models like AlexNet, ANN, RNN, etc. Furthermore, there are many other ways of diagnosing the tumor like a biopsy, MRI, etc for which different datasets can also be considered.

The model performed comparatively less accurately with VGG, ResNet, and DenseNet architecture. Moreover, the size of the dataset chosen was comparatively smaller and can be increased in the future to provide an appropriate number of images for both training and testing the model.

9. Conclusion

Through this experimentation, the model with a dropout rate of 0.1 with a patch size of 50 x 50 demonstrated considerably better accuracy than other models. These models can also be used to predict the aggressiveness of a tumor, which can aid in treatment planning. However, it is important to note that while these models have shown moderate results, they are still being researched and more studies are needed to validate their use in clinical practice. Additionally, it is important to ensure that the data used for the training and testing of these models are diverse and representative of the population to avoid bias. Once the appropriate dataset has been compiled and pre-processed, various machine learning algorithms can also be applied to it to make predictions. The algorithm's selection depends on the task's specific requirements and the type of data being analysed.

In the future, various classification methods and optimization techniques can be used along with improved feature selection and image cleaning. Moreover, various experimentations can be performed over the architecture build to gain higher performance accuracy.

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